KNOTS CONJECTURES TILL Conjecture 20250815

We always talk about frictionless rope knots with a diameter of 1 and a minimum radius of gyration of 1 (minimum distance from the neutral fiber to itself = 1). It's also understood that the rope is inextensible and frictionless.

I believe, like you, that since there are infinitely many knots, the minimum rope length (which certainly exists) might not have a unique spatial configuration for all knots. The trifolium-right and trifolium-left have the same. However, we can conjecture (why not?) that for alternating knots, the minimum chord length is unique for each knot (with the exception of L-R symmetries).

So, several conjectures:

20250712 Conjecture:

For all alternating open or closed knots, the minimum chord length is invariant, and the shape of the knot at that minimum is also invariant. For other types of knots, there could be a minimum length (there always is) with several different spatial configurations.

20240222 Keyhole Conjecture:

Let a wall have a 2x2 square opening. A chain of links of diameter 1 passes from one side of the wall to the other through that opening. For knots with a chord diameter of 1 and that we can extend or tighten:

any open (or closed) knot does not necessarily pass from one side of the wall to the other.

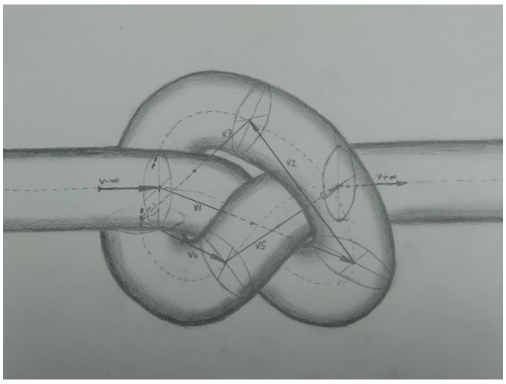
The trefoil knot passes, the figure-eight knot requires a 3x2 opening, etc.

20240606 Conjecture or Inscribed minor polygon:

For open knots (since they have a beginning and an end), a unique definition of a knot would be the shortest list of vectors (or line segments) that fits within the cylindrical body of the rope. See the image below to construct such vectors.

v1 runs from the beginning of the neutral fiber's curvature to itself, but is also tangent at a point to the cylindrical body of the rope.

v2 does the same, but starts at the tip of v1, and so on.



Conjecture 20250802A Computational Conjecture in Z3:

If we allow the neutral fiber of a knot (open or closed) to change direction only if it is in Z3 (see the Appendix for an image of the minimal trifolium in Z3, embedded in R3, of course), the Reidemeister operations become computationally immediate given the collision distance of 1.

I will call these knots **K(n) Polyg Z3.**

Note: with the program simulate-polymer.bin, these operations are immediate (in R3) almost always precisely because they are in R3 and we stretch. In molecular dynamics (MD), this programming precaution is called excluded volume calculation, in plain language: where there is already one monomer, there cannot be another at the same time.

Conjecture 20250802B Minimum ropelength conjecture for knots in a crystal lattice (K(n) Polyg Z3):

Given the previous condition with polygonal knot: the neutral fiber can only change direction if it is in Z3, that is, the position vector in the turns are three integers (p, q, r), the minimum length configuration is no longer unique (see Appendix at the end).

Conjecture 20250802C or Equivalence conjecture:

K(n) (in R3) is equivalent to the polygonal K(n) Polyg Z3embedded in R3.

Note: This conjecture/axiom could be eliminated, but it would be helpful to first examine what happens with the two K_8_17 knots.

Conjecture 20250813 or Closeness Conjecture: The minimum ropelength (K(n)) in R3 for alternating knots is reached from some equivalent minimum of K(n) Polig Z3 by removing the tie to Z3 and, of course, stretching.

The sum of the vector product of all the displacement vectors (p **i**, q **j**, r **k**) two by two on each side of the knotted polygon is sometimes different.

Since the rope thickness is 1, if we suddenly allow the neutral fiber to move away from the points in Z3, the knot will reach a minimum ropelength.

Since Conjecture 20250802B states that the configuration and ropelength of K(n) Polig Z3 are not unique, we just have to try stretching all of them in R3 and see the minimum. No osculating spheres are needed.

I don't know how to find all the K(n) Polig Z3 of K(n).

This is an interesting problem because let's say that K(n) passes from R3 to Z3, which is a different approach than the usual one: putting the entire knot K(n) between the plane z=0 and the plane z=1 (bridges).

Last Conjecture 20250815 or K(n) can be described as a tensor:

Each K(n) can be described as a Z3xZ3x---Z3 (m times) tensor or as least as a tensor.

Note: If we consider each knot as a taut cylindrical rope, then the forces on its cylindrical surface can be viewed as a force field or tensor field.

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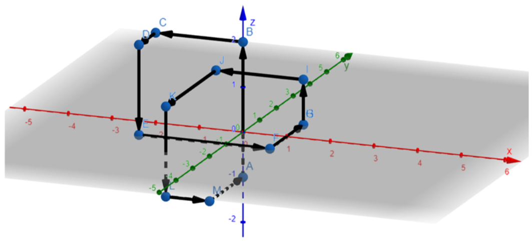
APPENDIX

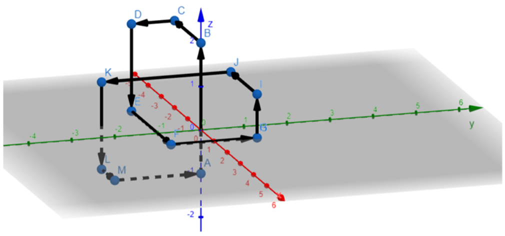
POLYGONAL TRIFOLIUM IN Z3

Trifolium 12 points (PMendivil).

Geogebra visualizes:

(This knot is, let's say, K3LL) other view:





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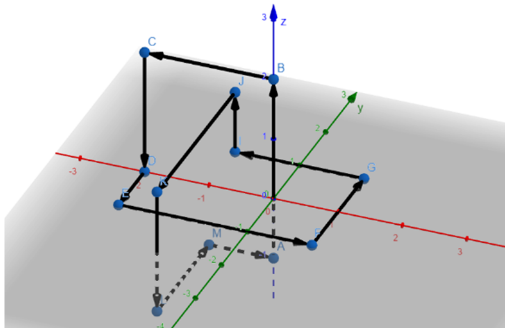
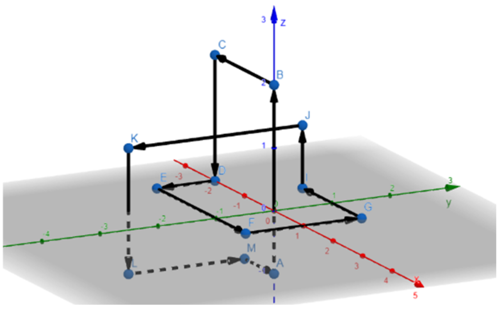
This is a minimal Z3 Trifolium (L) but is not unique:

(This knot is, let's say, K3LR)

other view:





Knot K3LL:

Pos	Point	Delta L	Along axis
A	(0, 0, -1)		
B	(0, 0, 2)	3	z
C	(-2, 0, 2)	2	x
D	(-2,-1, 2)	1	y
E	(-2, -1, 0)	2	z
F	(1, -1, 0)	3	x
G	(1, 1, 0)	2	y
I	(1, 1, 1)	1	z
J	(-1, 1, 1)	2	x
K	(-1, -2, 1)	3	y
L	(-1, -2, -1)	2	z
M	(0, -2, -1)	1	x
A	(0, 0, -1)	2	y

Knot K3LR:

K3 path 2:

Pos	Point	Delta L	Along axis
A	(0, 0, -1)		
B	(0, 0, 2)	3	z
C	(-2, 0,2)	2	x
D	(-2,0, 0)	2	y
E	(-2, -1, 0)	1	z
F	(1, -1, 0)	3	x
G	(1, 1, 0)	2	y
I	(-1, 1, 0)	2	z
J	(-1, 1, 1)	1	x
K	(-1, -2, 1)	3	y
L	(-1, -2, -1)	2	z
M	(-1, 0, -1)	2	x
A	(0, 0, -1)	1	y

For K_3_LL: ∑ (v1xv2, v2xv3... v11xv12) = (14, -14, 14)

For K_3_LR: ∑ (v1xv2, v2xv3... v11xv12) = (10, -10, 10)